

Physics necessary for human space travel to Mars

A manned Mars mission has captured the imagination of humanity since Soviet cosmonaut Yuri Alekseyevich Gagarin became the first person to travel into space in 1961 [1]. During the late 20th century many mission concepts for travel to Mars were developed. The first engineering analysis of a crewed Mars mission was written by Werner von Braun [2], who helped develop the V-2 rocket during World War II and later served as the chief architect of the Saturn V launch vehicle that propelled the Apollo spacecraft to the moon. The proposal was published as Das Marsprojekt in West Germany in 1952 with an expected launch year of 1965 and consisted of a 'flotilla' of ten 4,000-ton ships with 70 crew members [3]. Since then, the exploration of Mars has been the goal of many space agencies. Plans vary from scientific expeditions consisting of a small group of astronauts, to the permanent colonization of Mars. The Mariner 4 Spacecraft performed the first successful Mars flyby in 1965, its closest approach was 9,846km, during which it returned valuable scientific data as well as the first close-up pictures of the Martian surface [5]. The data gathered from the flyby showed a surface atmospheric pressure less than 1% of Earth's and daytime temperatures of -100 degrees celsius, meaning Mars was less hospitable to life than previously thought. The first craft to land on the surface of Mars was the Mars 3 space probe, attaining a soft-landing in 1971. However the lander failed 14.5 seconds after landing, without transmitting any data [6]. Viking 1 became the first spacecraft to land on Mars and successfully perform its mission, transmitting data from the surface for 2307 days [7]. Surveys carried out by later orbiters have mapped Mars in higher resolution with improved instruments and determined the most probable locations of subsurface water.

Many private organisations and national space agencies have current plans to launch a manned expedition to Mars, with SpaceX setting an ambitious target of 2024 for its first manned flight to Mars [8]. NASA is also planning a manned mission with a target launch year of 2033 [9]. Nasa is currently working on a deep space rocket known as the Space Launch System (SLS). The core stage of the rocket will be 64.6m in length and 8.4m in diameter, and will be powered by four RS-25 engines, currently being built by The Boeing Company. The RS-25 engines use cryogenic liquid hydrogen and liquid oxygen as fuel. The block 2 configuration of the rocket will be designed to provide 5.4 million kilograms of thrust, capable of lifting 45 tonnes to deep space [10]. The SLS is designed to lift an Orion crew module, capable of supporting a crew of 2-6 astronauts. The crew module uses the Avcoat ablator system for its heat shield. Avcoat is composed of silica fibers (liquid glass) embedded in a fibreglass honeycomb matrix [11], and has been tested in temperatures as high as 5000 degrees Fahrenheit, allowing it to re-enter the Earth's atmosphere [12]. For a Mars mission the Orion capsule would be paired with a habitation module, which is planned to be 8.4 m in diameter and 11.7 m in length and have an attached laboratory with research instrumentation for physical sciences [13]. The NASA Mars mission also makes use of a proposed space station known as the Lunar Orbital Platform-Gateway, which is intended to be placed in a highly elliptical orbit around the moon, bringing the station within 3,000 km of the lunar surface at closest approach and as far away as 70,000 km

[14]. The station would offer the opportunity for a gradual buildup of resources and capabilities over time for a manned Mars mission.

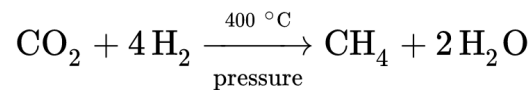
The proposed SpaceX manned Mars mission consists of a two stage rocket system known as the Big Falcon Rocket (BFR), with a second stage known as the Starship that will be used as a spacecraft to reach Mars and return to Earth [15]. To allow a larger payload to reach Mars the Starship would undergo refilling of propellants in Earth orbit before departing to Mars. This would be achieved by initially launching a tanker carrying propellant to Earth orbit. The first stage of the SpaceX rocket is known as the Super Heavy, expected to be 63 m long and 9 m in diameter. This stage will be powered by 31 Raptor engines, providing 6.3 million kilograms of thrust [16]. The Raptor engines use cryogenic liquid methane and oxygen as fuel, the use of subcooled propellants increases propellant density allowing more mass to fit in propellant tanks, subcooled propellants also increase engine performance [17]. The second stage of the rocket, Starship, is 55m long and 9m in diameter, constructed using stainless steel [18]. The Starship will have a pressurized volume of 1000m³, which could be configured for up to 40 cabins [19]. Starship will be powered by six Raptor engines, three optimized for sea level and three optimized for vacuum, giving a total thrust of 1.2 million kilograms [20]. The Starship prototype made its initial flight test in July 2019, with the first orbital test planned for 2020 [21]. Both stages of the rocket are designed to be fully reusable, capable of reentering Earth's atmosphere and landing vertically.

The longer term SpaceX plan for human travel to Mars involves establishing an extended surface presence. The first stage of the plan involves sending two robotic cargo flights in 2022, carrying an array of solar panels, mining equipment, surface vehicles and life support infrastructure [22]. Four more Starships would follow in 2024, two more cargo flights and two crewed flights, with the aim of setting up a propellant producing plant [23]. The plant would produce methane and liquid oxygen from subsurface water ice and atmospheric CO₂. The first temporary habitats on the surface would be the Starships themselves, as they have their own life support systems. The proposed landing zone would be located at less than 40° latitude for optimized solar power production and a relatively warm temperature. The landing zone must also be near a sub-surface water ice deposit to allow propellant production [24].

The most likely basis for propellant production is the Sabatier methanation reaction [25]. This reaction can exploit the availability of carbon dioxide in the Martian atmosphere, around 95.3%, to create both methane fuel and water. Although the atmospheric pressure on Mars is less than one hundredth of Earth's, the abundance of carbon dioxide means the Martian concentration is almost 20 times as much carbon dioxide as is in an equivalent amount of the Earth's atmosphere. However the martian atmosphere also consists of 2.7% nitrogen and 1.6% argon as well as trace amounts of other gases, including oxygen and water [26]. This means a gas separation is needed before the Sabatier process can occur. The most common method on Earth for the separation of air into its component gases is the use of a Carbon Molecular Sieve (CMS) [27], however with high costs of transport for the materials for a CMS from earth, it may be more practical to use a multi-stage chiller pressure cycle subsystem to separate the Martian

atmospheric constituents. This involves cooling the martian air until each gas condenses into a liquid, which is then extracted [28]. Although this process uses more energy than a CMS it could be more efficient due to high transport costs.

The Sabatier process involves the reaction of hydrogen with carbon dioxide at elevated pressures and temperatures (optimally 300-400°C), in the presence of a catalyst. Nickel or, more optimally, aluminium oxide acts as the catalyst [29]. It is described by the exothermic reaction,



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The Sabatier process requires a source of free hydrogen to react with the carbon dioxide to produce methane. The water produced as a byproduct of the reaction can act as a partial source of hydrogen through electrolysis, however not enough is produced to function as the sole source [30]. A preliminary study by SpaceX estimates that a ton of water ice per day must be mined and its impurities filtered to provide enough hydrogen for the propellant plant. Based on a 2011 prototype test operation, one ton of propellant uses around 17 megawatt-hours energy input from solar power. The total energy needed to produce a full load of propellant for a BFR rocket is around 16 gigawatt hours [23]. To produce the energy needed in 26 months would require just under one megawatt of continuous electrical power. A ground based array of thin film solar panels would need to have an area of 56,200 square metres to produce sufficient power, this array would have a mass that could be transported by a single BFR [24].

The energy needed for transfer between Earth orbit and Mars orbit is lowest at intervals fixed by the synodic period. This minimum occurs every 26 months, so missions are typically planned to coincide with one of the launch windows. Due to the eccentricity of Mars' orbit, the energy needed also varies on a 15 year cycle, with the lowest energy windows needing only half the energy of the peaks. The next low-energy launch window occurs in 2033 [31]. The total travel time for a round trip takes depends on the flight route taken. Several mission types have been proposed, including opposition and conjunction class. Conjunction-class missions are characterized by minimal total energy requirements and extended mission duration. Opposition class mission include greater change in velocity maneuvers that result in shorter mission durations [32]. Conjunction class missions have generally been the focus of spaceflight to this date, however opposition-class concepts could play a critical role in the development of an integrated campaign for human exploration of Mars. The health risks for initial human Mars missions will be high, as the duration that humans spend in space will be stretched far beyond what has been presently accomplished. Conjunction-class trajectories benefit from the planetary alignment of Earth and Mars for both the departure and return legs of the mission, as shown in figure 1. This results in short transit durations and long durations in Martian space, typically around 600 days, resulting in a total mission duration of around 1000 days [33].

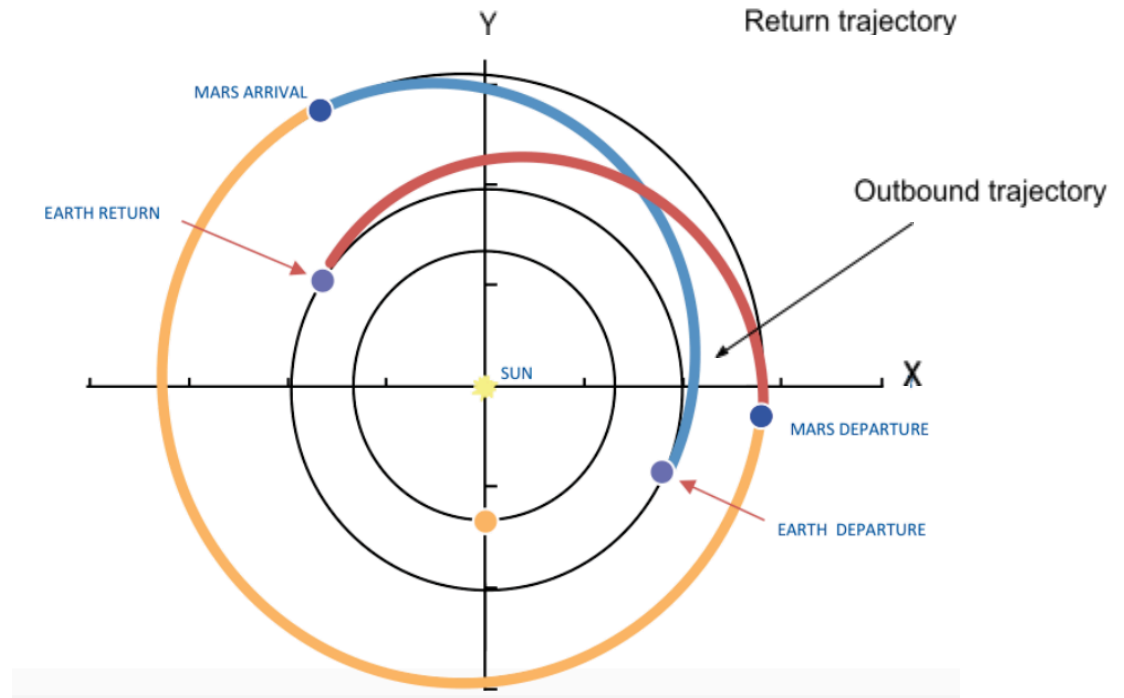


Fig 1. Example of a conjunction trajectory [33]

Opposition class missions will utilize an optimized trajectory for the outbound leg but will then use a higher-energy trajectory for the return leg, resulting in shorter durations in Martian space, typically around 50 days, resulting in a total mission time of 550 days [33]. Some opposition-class trajectories use the gravitational field of Venus as a propelling force by travelling around the planet, reducing the journey time back to Earth as shown in figure 2. This is known as the Crocco flyby and was proposed by Gaetano Arturo Crocco in 1965 [34].

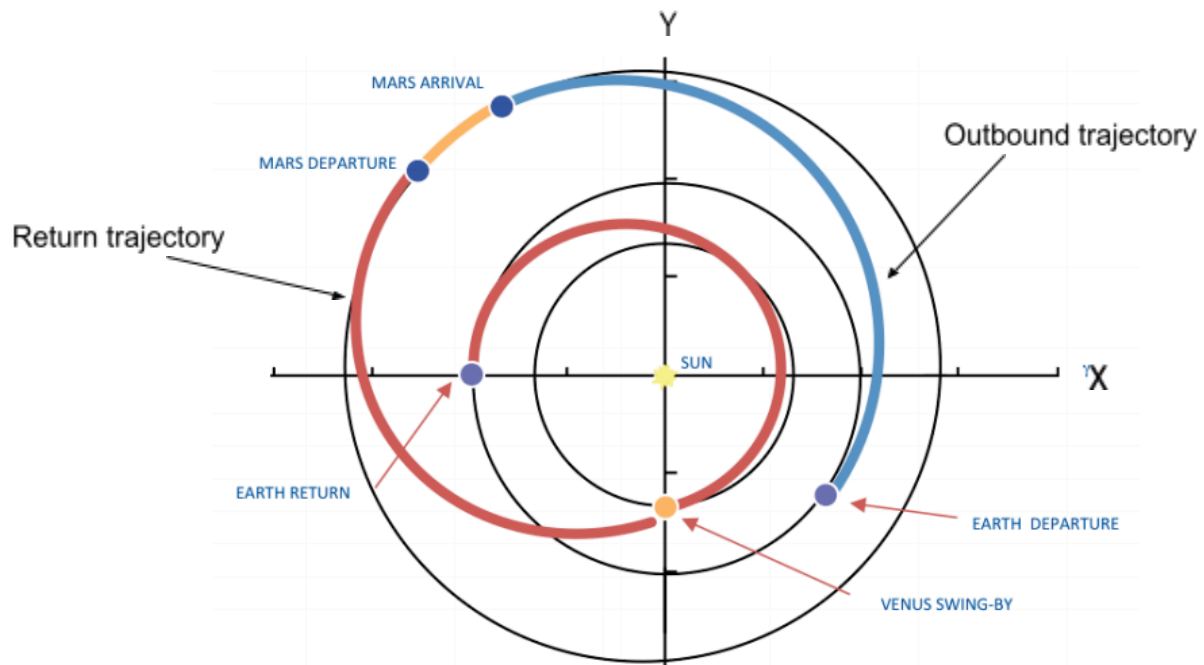


Fig 2. Example of an opposition trajectory, including a Crocco flyby [33]

The lowest energy transfer to Mars is a Hohmann transfer orbit, involving a 9 month travel time from Earth to Mars, approximately 500 days in Martian space and 9 months back to Earth. The orbital maneuver to perform the Hohmann transfer uses two engine impulses, one to move a spacecraft onto the transfer orbit and a second to move off it, this is shown in fig 3.. The transfer is based on two instantaneous velocity changes. Extra fuel is required to compensate for the fact that the bursts take time; this is minimized by using high-thrust engines to minimize the duration of the bursts [35].

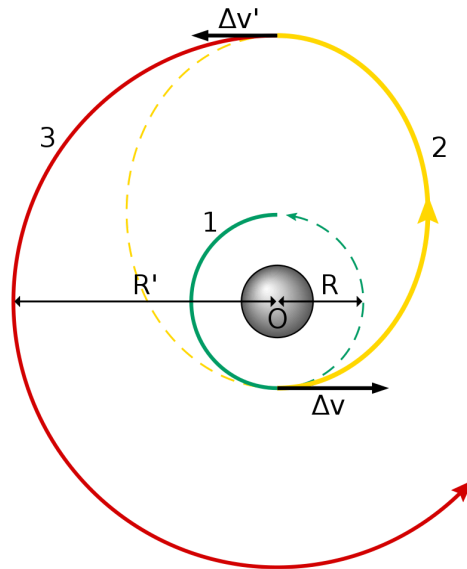


Fig 3. Hohmann Transfer orbit (2), from an orbit (1) with radius R , to a higher orbit (2) with radius R' . Two instantaneous velocity changes ΔV and $\Delta V'$ are needed [35]

A major obstacle facing any manned mars missions are the physical challenges of spending over 500 days in space. There are major health threats from the flux of high-energy galactic cosmic rays (GCRs), along with radiation from solar proton events (SPE's) and radiation belts. GCRs consists of high energy protons (85%), helium (14%) and other high energy nuclei [36]. Heavy ions and low energy protons are highly ionizing forms of radiation, which produce distinct biological damage compared to gamma and X-rays. Since a manned flight of this length has not been attempted, no human epidemiology data for cancer and other fatal risks associated with GCRs exist. The Mars Science Laboratory travelled from Earth to Mars in 2011-2012, measuring radiation inside the spacecraft using the Radiation Assessment Detector (RAD). The calculated radiation dose was 0.66 sieverts for the round trip. NASA's career radiation limit for astronauts is 1 sievert [37]. There are also adverse health effects associated with prolonged weightlessness, including muscle atrophy and deterioration of the skeleton. Other significant effects include a slowing of cardiovascular system functions and decreased production of red blood cells. Spaceflight induced visual impairment is another risk, with NASA reporting 15 astronauts experiencing visual changes during or after long-duration flights [38]. This is hypothesized to be a result of increased intracranial pressure. The physiological effects of isolation from Earth, as well as the social effects of several people living in cramped conditions for over a year must be taken into account.

Overall the largest limiting factor for launching a manned Mars mission is funding. The estimated cost of an expedition is roughly 500 billion U.S. dollars [39]. Critics argue that the immediate benefits of establishing a human presence on Mars is outweighed by the financial cost of the endeavour, and that funds could be better redirected towards ongoing problems on Earth. Proponents of human space exploration contend that the symbolism of setting foot on another planet will garner public interest and encourage global cooperation.

Word count: 2312

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